AI-Driven Examination Analysis

Using LLMs and RAG Architecture for Syllabus Alignment

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Executive Summary

- LLM-based exam analysis pipeline to evaluate 2025 Geography Paper against syllabus requirements
- The system checks: 1) syllabus alignment, 2) topic weightage, 3) assessment objective (AO1/2/3) coverage
- Architecture uses a LLM and RAG Framework
  - Multi-Language Retrieval-Augmented Generation (RAG)
  - State-of-art LLM Reasoning
- Scale to multiple subjects, languages, and exam formats

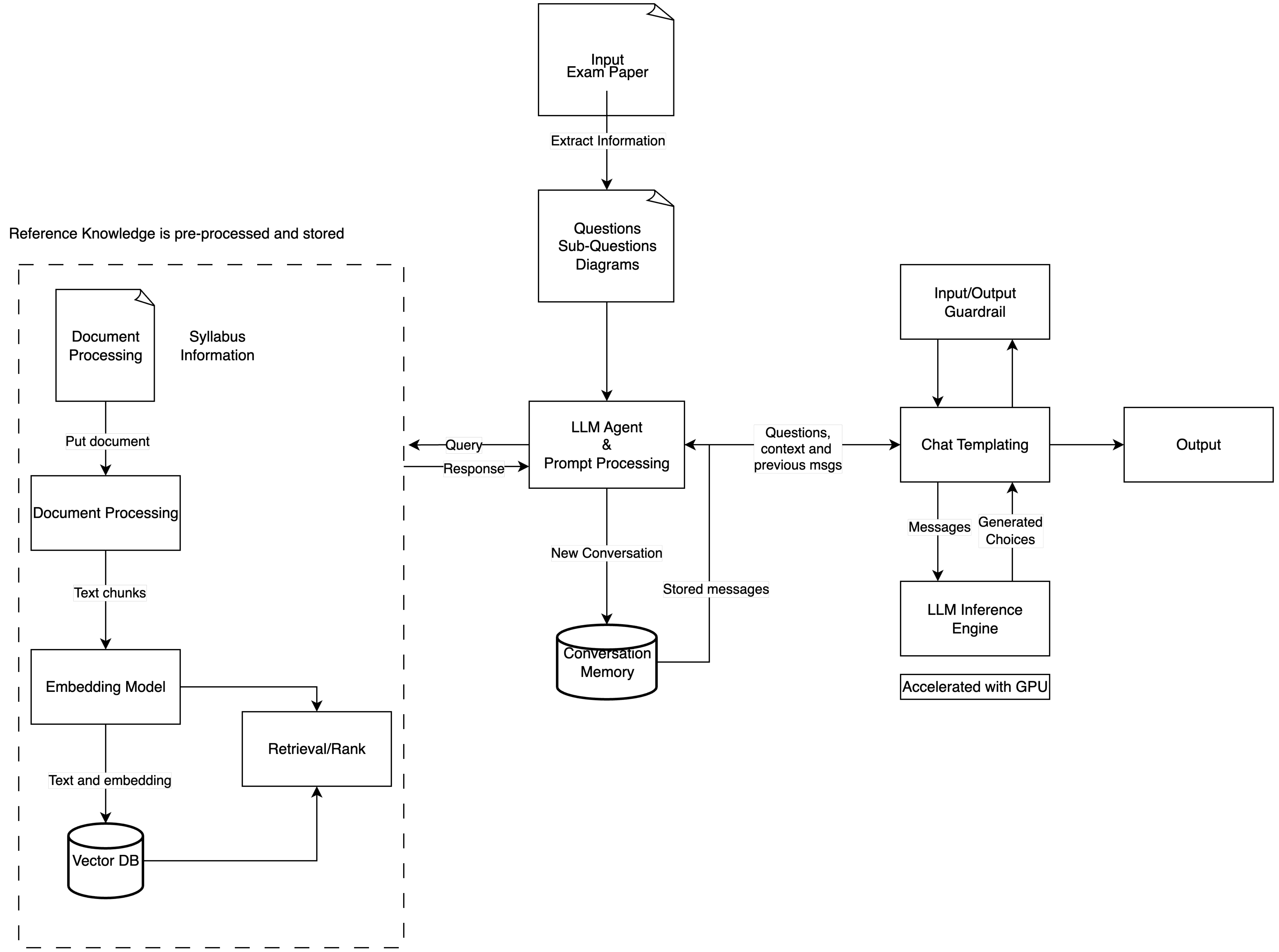
Problem Statement

"How can AI automatically review an exam paper for syllabus alignment, topic balance, and AO distribution?"

We propose a solution that:

- Reads exam PDFs + syllabus PDFs
- Understands diagrams, maps, structured questions
- Assigns topics, clusters, AOs
- Computes coverage & generates reports
- Works for Geography today, any subject tomorrow

Pipeline Overview



Component	**Description**	**Key Technologies**
Exam Paper Input	User uploads the 2025 Geography exam PDF	File uploader/Storage
Extract Information	Parse PDF into questions, sub-questions, marks into structured content for syllabus matching	PyMuPDF, PDFPlumber, Unstructured
OCR & Diagram Parsing	Reads maps, diagrams, tables using multimodal LLMs to convert non-text into evaluable text	GPT4o, Vision LLM, OCR
Document Chunking Engine	Split structured information into chunkable elements for processing	Text splitters, text-embedding
Vector Database	Stores syllabus representations for high-speed lookup and alignment	ElasticSearch, Milvus, Pinecone, FAISS, ChromaDB
Retrieval/Rank Layer	Ensures correct syllabus chunk is matched to each exam question	Bee-ranker, ColBERT
LLM Agent & Prompt Processor	Core reasoning unit: topic tagging, AO classification, alignment judgement	GPT, Claude, Gemini
Conversation Memory	Stores reasoning context and state for multi-turn analysis and consistency	Redis, LangChain memory
Input/Output Guardrail	Detect hallucinations, enforce citations, validate safety	NeMoGuardrails, LlamaGuard,
Chat Templating	Prompt scaffolding for consistent reasoning	LangChain, PromptTemplate
LLM Inference Engine	Executes reasoning tasks at scale	GPU-accelerated servers (A100, H100, L40s)
Output Generator	Creates alignment reports, charts, weightage tables	HTML, Markdowns

POC - AI Analysis

- Key Evaluation Goals
  - Syllabus Alignment - Does each exam question align with syllabus objectives?
  - Topic Weightage - Is the distribution of topics balanced across clusters?
- POC Pipeline Overview

Stage	**Description**	**Tools Used**
Data Extraction	Extract exam questions and syllabus	PyMuPDF, OCR Fallback
Preprocessing, Chunking, Embedding Generation	Breaks texts into vector embeddings	GPT4o, text-emedding-3-small
Retrieval (Vector Search)	Identify top-k relevant syllabus segments per question	FAISS
LLM Alignment Evaluation	Evaluate cluster/topic/AO alignment with scores	Chat Prompts

POC - AI Analysis

- LLM Prompting & Evaluation Strategy

Component	**Purpose**	**LLM Prompting/Logic**
Retrieval Prompt	Fetch relevant syllabus chunks	“Return top 5 syllabus chunks most similar to the question”
Alignment Prompt	Evaluate cluster/topic/AO alignment	Structured JSON prompt requesting: Cluster, Topics, A01/AO2/AO3, Score, Strength, Explanation
Alignment Scoring Rubric	Quantify alignment strength	0 - 5 scale (None → Very Strong) based on explicit syllabus match
Topic Classification Prompt	Assign question to specific topic(s)	Identify exact topics and explain why”
Weightage Analysis	Compute paper-level coverage	Python aggregation, no LLM required
Final Report Generation Prompt	Produce formatted summaries	Summarize alignment patterns and insights

POC - AI Analysis

- Illustration Outputs - Per Question Syllabus Alignment Results (in markdown format)

Results

Geography LLM Alignment & Topic Weightage Report

1. Per-Question Alignment Overview

Question 1

Question (snippet): 1 Cluster 1: Geography in Everyday Life A group of students investigated the experience of visitors at the Gallop Extension in the Singapore Botanic Gardens. The Gallop Extension is an eight-hectare area with many featur...

- **Matched Clusters:** Cluster 1: Geography in Everyday Life
- **Matched Topics:** Cluster 1 Topic 1.1: Geographical Methods
- **Assessment Objectives (AOs):** AO1, AO2
- **Alignment Score:** 5.0 (Strong)
- **Explanation:** The question aligns with Cluster 1: Geography in Everyday Life, specifically focusing on geographical methods such as sampling and questionnaire design. It tests students' understanding of data collection methods and their application in a real-world context, which is a key aspect of Topic 1.1.

Question 2

Question (snippet): 2 Cluster 2: Tourism (a) The growth of tourism results from the interaction between a range of factors. (i) Table 2.1 shows changes in household disposable income and the number of tourist departures from a developed cou...

- **Matched Clusters:** Cluster 2: Tourism
- **Matched Topics:** Cluster 2 Topic 2.1: What is tourism and impacts of tourism?, Cluster 2 Topic 2.2: What led to the growth of tourism?, Cluster 2 Topic 2.3: How does tourism affect the economies of places?
- **Assessment Objectives (AOs):** AO1, AO2
- **Alignment Score:** 5.0 (Strong)
- **Explanation:** The question aligns strongly with the syllabus by addressing the growth of tourism through factors like disposable income and mobility, as well as the economic and environmental impacts of tourism. It tests knowledge and understanding of tourism systems and their impacts, as well as skills in data interpretation and graphical representation.

Question 3

Question (snippet): 3 Cluster 3: Climate (a) (i) Describe the differences in rainfall between tropical equatorial and cool temperate climates. [2] (ii) Explain why rainfall and temperature differ between tropical equatorial and cool tempe...

- **Matched Clusters:** Cluster 3: Climate
- **Matched Topics:** Cluster 3 Topic 3.1: Weather and Climate, Cluster 3 Topic 3.2: How do anthropogenic factors contribute to climate change?, Cluster 3 Topic 3.3: How might climate change affect natural systems?, Cluster 3 Topic 3.4: How effective are adaptation strategies in building a community's resilience to climate change?
- **Assessment Objectives (AOs):** AO1, AO2, AO3
- **Alignment Score:** 5.0 (Strong)
- **Explanation:** The question aligns strongly with Cluster 3: Climate, specifically addressing differences in climate types (Topic 3.1), anthropogenic factors affecting climate (Topic 3.2), impacts of climate change (Topic 3.3), and the effectiveness of international agreements as a mitigation strategy (Topic 3.4). It tests AO1 through knowledge of climate differences, AO2 through explanation of these differences, and AO3 through evaluation of international agreements versus local initiatives.

👉 All questions show strong alignment to syllabus clusters and topics (scores 4 - 5)

POC - AI Analysis

- Illustration Outputs - Topic & Cluster Weightage Summary (in markdown format)

Results

2. Topic & Cluster Weightage Summary

Total questions analysed: **3**

2.1 Cluster Weightage

Cluster	Count	Proportion
Cluster 1: Geography in Everyday Life	1	0.33
Cluster 2: Tourism	1	0.33
Cluster 3: Climate	1	0.33

2.2 Topic Weightage

Topic	Count	Proportion
Cluster 1 Topic 1.1: Geographical Methods	1	0.33
Cluster 2 Topic 2.1: What is tourism and impacts of tourism?	1	0.33
Cluster 2 Topic 2.2: What led to the growth of tourism?	1	0.33
Cluster 2 Topic 2.3: How does tourism affect the economies of places?	1	0.33
Cluster 3 Topic 3.1: Weather and Climate	1	0.33
Cluster 3 Topic 3.2: How do anthropogenic factors contribute to climate change?	1	0.33
Cluster 3 Topic 3.3: How might climate change affect natural systems?	1	0.33
Cluster 3 Topic 3.4: How effective are adaptation strategies in building a community's resilience to climate change?	1	0.33

2.3 Assessment Objectives Distribution

AO	Count
AO1	3
AO2	3
AO3	1

👉 Paper evenly covers clusters 1, 2, 3 - aligned with syllabus intentions

Automated Evaluation Framework

- Automated Evaluation Framework for Syllabus Alignment & Topic Weightage Analysis
- Why automate? 👉 repeatable & objective evaluation, scalable scoring, hallucination safeguards, quality monitoring

RAG Retrieval Quality Evaluation		
Metric	Description	Purpose
Precision@k	Measures how many retrieved chunks are relevant using LLM scoring	Filters noisy or irrelevant syllabus sections
Hallucination Detection Module		
Faithfulness check	Does the answer contain statements NOT supported by the retrieved syllabus chunks? Answer: yes/no + probability.	Hallucination probability (0 - 1)
Contradiction check	Natural Language Inference check	No contradiction/Contradiction detected
Source-grounding scoring	Model must cite retrieved syllabus text using quotes	Faithfulness score
Attribution score	% of answer sentences supported by syllabus	Attribution score
LLM Reasoning Quality Evaluation		
Rudric adherence	Compare output to checklist derived from subject rubrics	% adherence
Consistency score	Re-run same prompt 3 times and measure variance	Low/Medium/High

Cross-Subject Adaptability

- How well these prompting techniques adapt across subjects?

Subject Type	Key Characteristics	Why current pipeline works?	Additional Adaptations Needed
Humanities (Geography, History, Social Studies)	Long-form responses, argumentation, source-based analysis	Retrieval + rubric-based prompting aligns naturally with content	Minimal: Update syllabus taxonomy & AO rubrics
Sciences (Physics, Chemistry, Biology)	Definitions, theory, formula manipulation, structured answers	RAG grounding ensures factual correctness	Add formula parsing, LaTeX recognition, unit consistency checks, diagram interpretation (vision)
Mathematics	Symbolic reasoning, calculation steps, proofs, graphs	RAG grounding ensures factual correctness	Numerical correctness solver, step-wise verification, graph extraction from images
English	Comprehension, summary, argumentative writing, linguistic features	Rubric-based prompting matches well with writing & comprehension rubrics	Add grammar scoring, coherence/logic scoring, stylistic feature detectors
Mother Tongue Languages (Chinese, Malay, Tamil)	Non-English	With multilingual models, the RAG → syllabus mapping generalizes well	Add native-language tokenization, cultural context awareness, tone/style scoring, other LLMs i.e., DeepSeek
Foreign Languages	Grammar, translation, communication tasks	RAG aligns questions with CEFR grammar/skill descriptors	Stronger multilingual embeddings and syntax-aware checks needed
Art	Visual analysis, interpretation, art history references, design processes	RAG retrieves syllabus context; AO mappings (perceive → analyse → create) work well	Add image processing (vision models) for artwork analysis
Computing	Algorithms, programming, logic, data structures	RAG retrieves correct programming constructs; AO mapping aligns with problem-solving hierarchy	Add code execution/sandboxing, static analysis for correctness
Physical Education/Sports	Physiology concepts	Syllabus retrieval supports correct topic mapping	Limited: Add diagram analysis (e.g., muscle group diagrams).

Conclusion

Summary of LLM and RAG pipeline

- POC pipeline was developed to analyze 2025 O-Level Geography Examination Paper, focusing on:
  - RAG retrieval using FAISS-grounded syllabus text
  - Syllabus alignment (clusters, topics, assessment objective)
  - Topic weightage

Evaluation of Adaptability

- Humanities subjects require minimal adaptation
- STEM and technical subjects require symbolic math parsing, diagram/image interpretation, code execution
- Non-English subjects require multilingual embeddings, culture context grounding, native-language tokenization